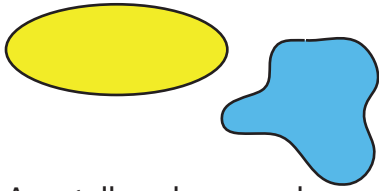


Progression on Area -- a visual overview, page 1 By Sybilla Beckmann Grade 3

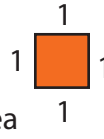
What is area?



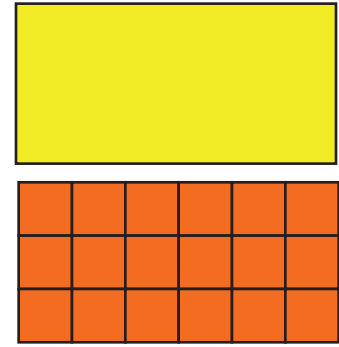
Area tells us how much "flat space" the shape takes up.

How many unit squares do we need to cover the rectangle without gaps or overlaps?

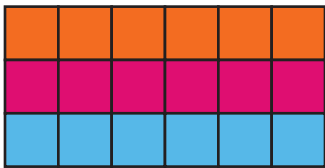
a unit square
1 square unit of area



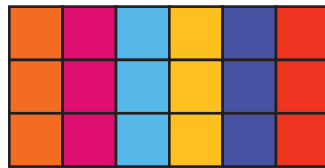
Area = 18 square units because it takes 18 unit squares to cover.



What's a quicker way than counting squares one by one?

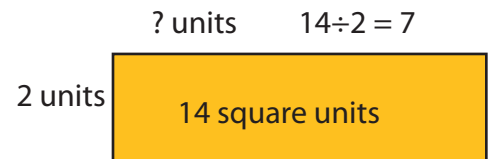
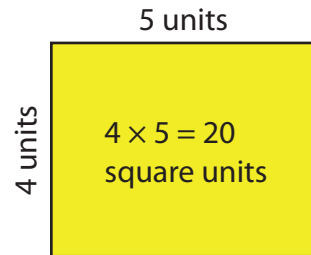


3 groups of 6 unit squares
 $3 \times 6 = 18$ square units

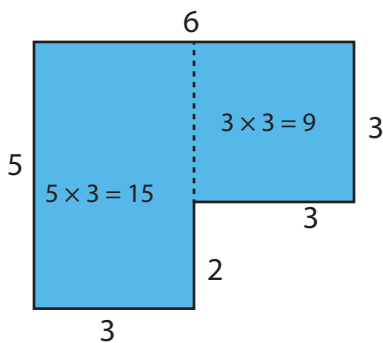


6 groups of 3 unit squares
 $6 \times 3 = 18$ square units

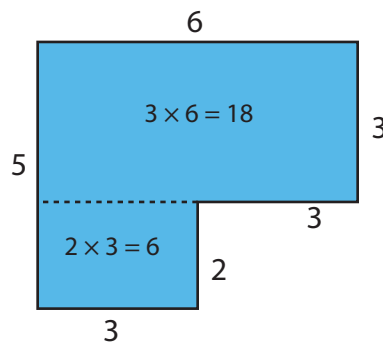
That is why we can multiply side lengths to find areas of rectangles.



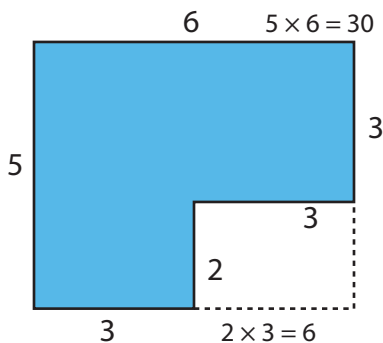
Decomposing into rectangles to find area



$15 + 9 = 24$
Area = 24 square units



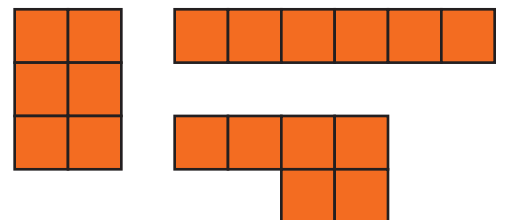
$18 + 6 = 24$
Area = 24 square units



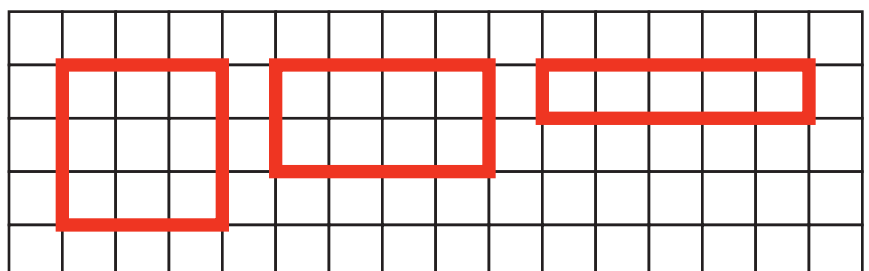
$30 - 6 = 24$
Area = 24 square units

Area is different from perimeter

Same area, different perimeter:

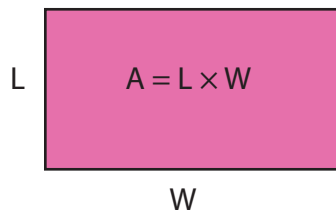


Same perimeter, different area:



Progression on Area -- a visual overview, page 2

Grade 4: apply area and perimeter formulas

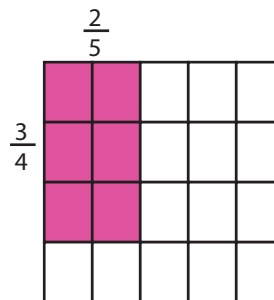
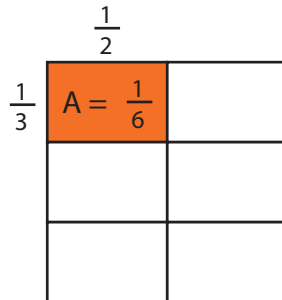


$$P = L + W + L + W \quad \text{or}$$

$$P = (2 \times L) + (2 \times W) \quad \text{or}$$

$$P = 2 \times (L + W)$$

Grade 5: Extend the area formula for rectangles to cases of fractional side lengths.



same answer!

$3 \times 2 = 6$ parts, so each is $\frac{1}{6}$ of a square unit.

$4 \times 5 = 20$ parts, so each is $\frac{1}{20}$ of a square unit. $3 \times 2 = 6$ parts shaded. So shaded area is $\frac{6}{20}$ square units.

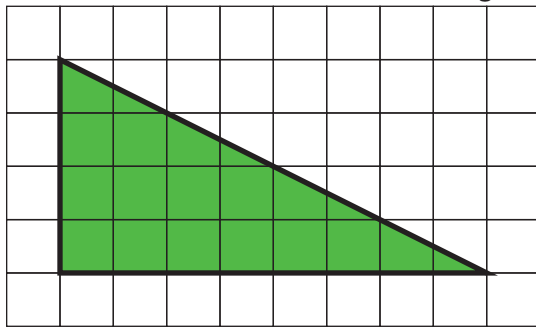
$$L \times W = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

$$L \times W = \frac{3}{4} \times \frac{2}{5} = \frac{6}{20}$$

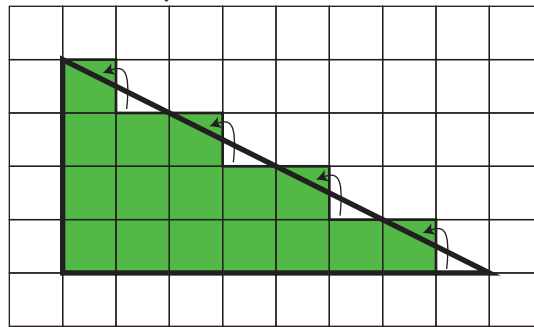
same answer!

Grade 6: Finding areas

What is the area of the shaded triangle?

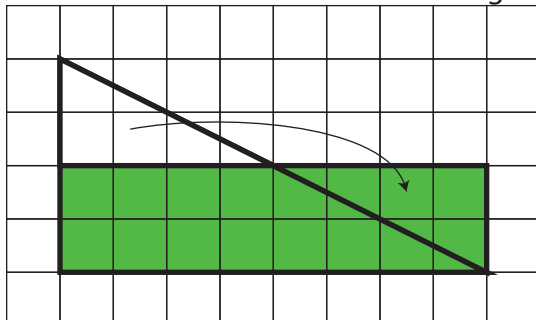


Most primitive method:

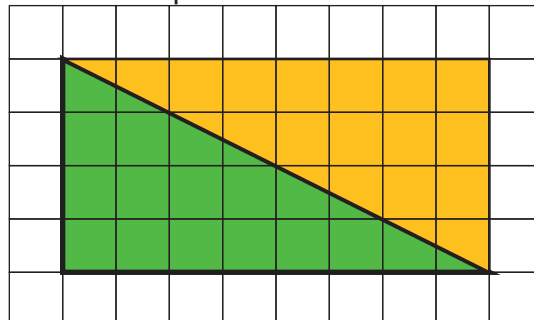


Move and combine small pieces; count the number of squares.

More advanced methods that will generalize to develop area formulas:



Move a chunk to create a rectangle of the same area.

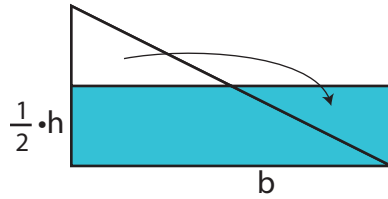
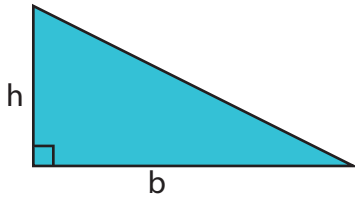


Combine two copies to make a rectangle of twice the area.

Progression on Area -- a visual overview, page 3

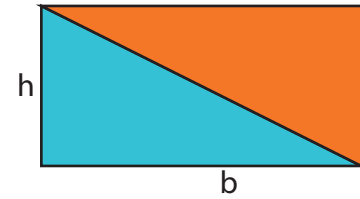
Grade 6: Developing area formulas.

1) Right triangles



$$A = b \cdot \frac{1}{2} \cdot h$$

two copies make a rectangle
of the same base and height

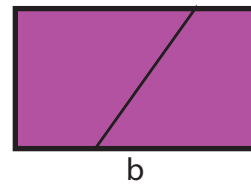
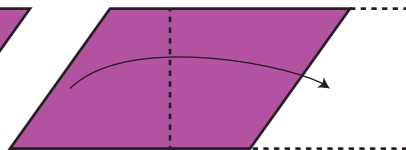
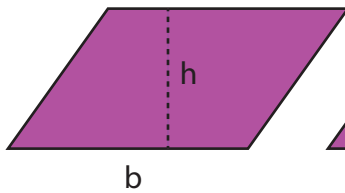


$$A = \frac{1}{2} \cdot b \cdot h$$

$$A = \frac{1}{2} \cdot b \cdot h \quad \text{or} \quad A = b \cdot h \div 2$$

Strategies:
Relate to a
rectangle
(or parallelogram)
by doubling then
halving, moving, or
adding then taking
away area.

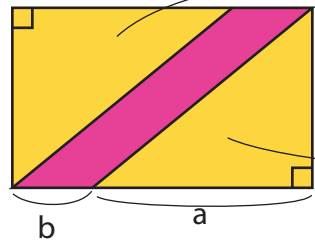
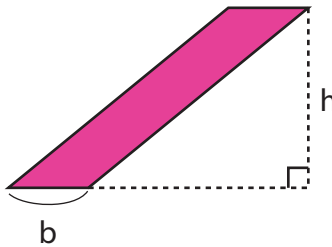
2) Parallelograms



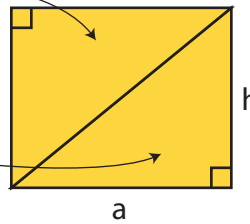
h

b

3) Very oblique parallelograms



$$\text{big rectangle's area} = (b + a) \cdot h = b \cdot h + a \cdot h$$

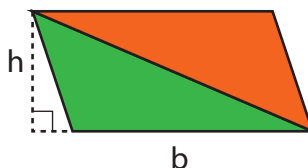
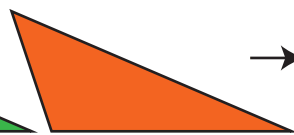
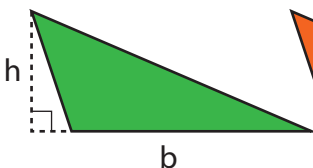


$$\text{two extra triangle's area} = a \cdot h$$

$$\begin{aligned} \text{parallelogram's area} &= (\text{big rectangle's area}) - (\text{two extra triangle's area}) \\ &= (b \cdot h + a \cdot h) - a \cdot h = b \cdot h \end{aligned}$$

$$A = b \cdot h$$

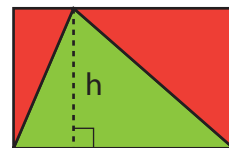
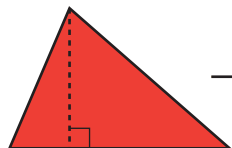
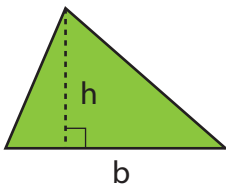
4) Other triangles: two copies make a parallelogram or rectangle of the same base and height, so the area of the triangle is half the area of the parallelogram or rectangle.



b

$$A = \frac{1}{2} \cdot b \cdot h$$

$$\text{or} \quad A = b \cdot h \div 2$$

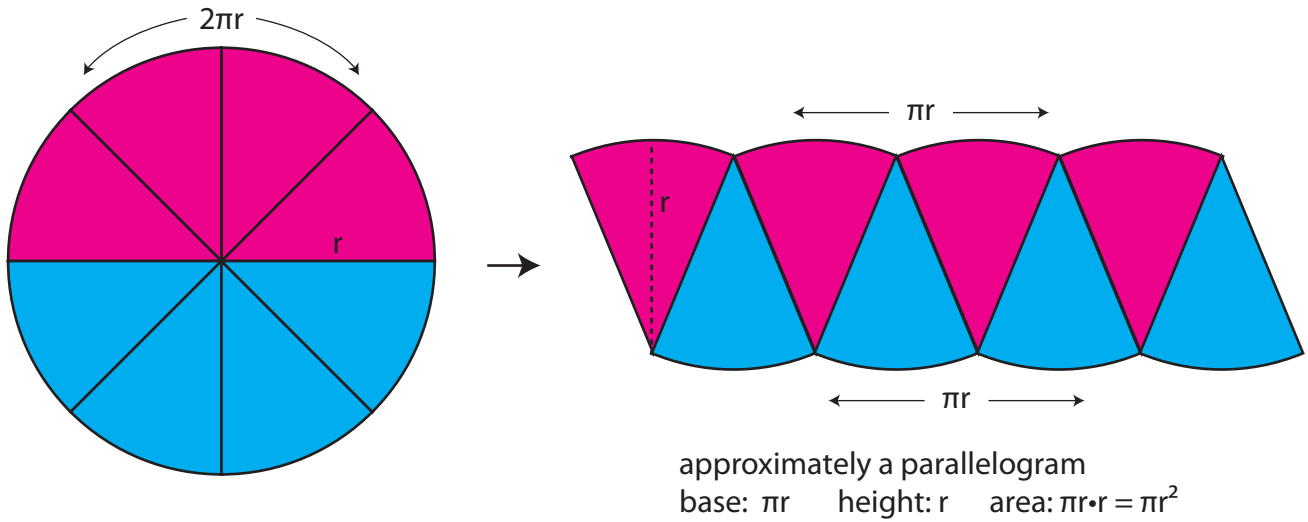


b

Progression on Area -- a visual overview, page 4

Decomposing and composing to relate areas.

Grade 7: Figure for an informal derivation of the relationship between the circumference and area of the circle.



Grade 8: Figure for a proof of the Pythagorean theorem.

